

# BTD Analytics Dashboard — Arsitektur Kerja Ekosistem UGM

**Repo payung:** <https://github.com/dedieko-priyadi/strategic-dashboard> **Tanggal:** 2026-08-07 | **Status:** LIVE

## Visi

Satu **main dashboard** yang menyintesis analytics dari SEMUA dataset BTD UGM — dengan sub-menu per dataset dan cross-domain analytics. Ini adalah **payung besar** project dashboard analytics BTD.

## Live URL

| Dashboard          | Public URL                                                                                                                      |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Strategic (PAYUNG) | <a href="https://nuc-nuc7i5bnh-1.tail758353.ts.net/strategic/">https://nuc-nuc7i5bnh-1.tail758353.ts.net/strategic/</a>         |
| EA Decision        | <a href="https://nuc-nuc7i5bnh-1.tail758353.ts.net/ea-decision/">https://nuc-nuc7i5bnh-1.tail758353.ts.net/ea-decision/</a>     |
| DSH Analytics      | <a href="https://nuc-nuc7i5bnh-1.tail758353.ts.net/dsh-analytics/">https://nuc-nuc7i5bnh-1.tail758353.ts.net/dsh-analytics/</a> |
| 9router            | <a href="https://nuc-nuc7i5bnh-1.tail758353.ts.net/9router-dash/">https://nuc-nuc7i5bnh-1.tail758353.ts.net/9router-dash/</a>   |
| Qdrant             | <a href="https://nuc-nuc7i5bnh-1.tail758353.ts.net/qdrant-dash/">https://nuc-nuc7i5bnh-1.tail758353.ts.net/qdrant-dash/</a>     |
| Wekan              | <a href="https://nuc-nuc7i5bnh-1.tail758353.ts.net/wekan-dash/">https://nuc-nuc7i5bnh-1.tail758353.ts.net/wekan-dash/</a>       |
| Charts Admin       | <a href="https://nuc-nuc7i5bnh-1.tail758353.ts.net/charts-admin/">https://nuc-nuc7i5bnh-1.tail758353.ts.net/charts-admin/</a>   |

## ARSITEKTUR KERJA — 7 LAYER

|                                                                                         |
|-----------------------------------------------------------------------------------------|
| L7 INFRASTRUKTUR: server, Docker, Tailscale, Git, Nextcloud, NAS                        |
| L6 KNOWLEDGE: BookStack, ea-ugm-explorer, shared-skills, ugm-strategic-analysis         |
| L5 DASHBOARD: strategic (payung) + 8 per-dataset                                        |
| L4 MCP: ugmcore, dsh, 9router, planka, wekan, regulasi — jembatan Hermes → data         |
| L3 CHATBOT/AI: ea-chatbot, dsh-chatbot, dsh-aitool, ugmcore-ai, LISA, openbtd           |
| L2 PLATFORM/WEB: dsh-ugm, ugmcore_main, planka, n8n, snapotter, sistemantrian, docuseal |

|                                                                 |
|-----------------------------------------------------------------|
| L1 DATA: eaugm_2025, ugm_dsh, 9router, Qdrant, Wekan, charts.db |
|-----------------------------------------------------------------|

## L1 — SUMBER DATA

Semua data diakses **read-only**, disimpan di **charts.db** (collection engine) via collector cron.

| Dataset | Lokasi                                        | Isi                                                                     | Kredensial                |
|---------|-----------------------------------------------|-------------------------------------------------------------------------|---------------------------|
| EA      | SQL Server eaugm_2025 @ 10.17.104.247         | 132.183 elemen, 6.151 diagram, 2.604 risiko, 47.925 proses, 481 unit    | sa / EA_SQL_PASS (config) |
| DSH     | MySQL ugm_dsh @ 10.17.104.219 (tunnel 13307)  | search_history 3.812, ai_logs 2.362, Q&A 338, feedback 41, index 89.774 | hermes_ro / H3rm3sR0#2024 |
| 9router | SQLite data.sqlite @ PC-AMD7900X 10.15.17.190 | usage_history 30K, api_keys 24, cost per model                          | SSH dedie                 |
| Qdrant  | localhost :8599 (NUC)                         | ea_elements 17K vectors, dsh_search                                     | API lokal                 |
| Wekan   | Wekan API                                     | board/cards project                                                     | token MCP                 |
| Planka  | Planka API 127.0.0.1:13370                    | kanban project                                                          | MCP planka                |

**PITFALL KRITIS:** DSH = ugm\_dsh BUKAN store\_ai ! EA = eaugm\_2025 . Jangan tertukar.

### Collector → charts.db (pipeline)

| Collector                 | Sumber         | Cron         | Tabel di charts.db                                                                                                                                                                    |
|---------------------------|----------------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| collect_ea_ugm.py         | SQL Server EA  | 04:00 harian | r9_ea_packages, r9_ea_elements (132K), r9_ea_connectors (94K), r9_ea_diagrams (6.151), r9_ea_diagram_objects                                                                          |
| collect_dsh.py            | MySQL ugm_dsh  | 04:30 harian | r9_dsh_search_history, r9_dsh_ai_logs, r9_dsh_qa_knowledge, r9_dsh_ai_feedback, r9_dsh_ai_conversations, r9_dsh_ai_analytics, r9_dsh_popular, r9_dsh_entity_trends, r9_dsh_facilities |
| collect_9router_detail.py | SQLite 9router | tiap jam     | r9_usage_history, r9_api_keys                                                                                                                                                         |

Semua dashboard baca charts.db **read-only** (volume mount :ro ).

## L2 — PLATFORM & WEB

| Project | Fungsi                             | Port        | Repo    |
|---------|------------------------------------|-------------|---------|
| dsh-ugm | Web DSH asli (PHP, 89.740 entitas) | :8585 /dsh/ | dsh-ugm |

| Project       | Fungsi                 | Port                 | Repo              |
|---------------|------------------------|----------------------|-------------------|
| ugmcore_main  | Web EA UGM             | :8580                | ugmcore_main      |
| planka        | Kanban modern          | 13370/13371 /planka/ | planka            |
| n8n           | Workflow automation    | :8550/8551           | n8n-magang        |
| snapshotter   | Transcribe audio/video | 13490 /snapshotter/  | snapshotter       |
| sistemantrian | Antrian layanan        | :8570                | sistemantrian-ugm |
| docuseal      | Digital signature      | :3000                | docuseal          |
| bookstack     | Knowledge base docs    | :8603                | bookstack-btd     |
| openbtd       | PoC chatbot publik BTd | :8600                | openbtd           |

## L3 — CHATBOT & AI

| Project        | Fungsi                                | Kanal                 | Repo        |
|----------------|---------------------------------------|-----------------------|-------------|
| ea-chatbot     | Chatbot EA (Hermes profile, MCP-only) | Telegram @ugm_dsh_bot | ea-chatbot  |
| ea-chat-ui     | Web UI chatbot EA                     | :3000                 | ea-chat-ui  |
| dsh-chatbot    | Chatbot DSH                           | :8516                 | dsh-chatbot |
| dsh-aitool     | AI tool (chat + generator berita)     | :8590                 | dsh-aitool  |
| ugmcore-ai/api | AI query EA (query-ai.ps1)            | :8580                 | ugmcore-ai  |
| LISA-UGM       | AI asisten (legacy)                   | —                     | LISA-UGM    |

**Pola:** Hermes profile + Telegram, public bot = toolsets mcp-only + branding hidden + TELEGRAM\_HOME\_CHANNEL.

## L4 — MCP SERVERS (jembatan Hermes → data)

| MCP      | Tools                                                                                    | Data            | Lokasi config                  |
|----------|------------------------------------------------------------------------------------------|-----------------|--------------------------------|
| ugmcore  | 11: test, summary, search, detail, proses-bisnis, aplikasi, search_qdrant, regulasi, dsh | EA eaugm_2025   | mcp_servers (command)          |
| dsh      | 7: facilities, news, faq, academic, publications, stats                                  | DSH ugm_dsh     | mcp_servers (URL)              |
| 9router  | 6: health, keys, usage, settings                                                         | 9router API     | mcp_servers                    |
| planka   | 13: project, board, list, card, user, member                                             | Planka API      | mcp_servers (command, mcp 1.x) |
| wekan    | 21: board, card, checklist, member, label                                                | Wekan API       | mcp_servers (command)          |
| regulasi | search_regulations                                                                       | Qdrant regulasi | mcp_servers (command)          |

**Config:** `mcp.servers` (URL-based) vs `mcp_servers` top-level (command-based stdio). **Pitfall:** mcp package == 1.x utk FastMCP (2.0 tidak punya `mcp.server.fastmcp`).

## L5 — DASHBOARD ANALYTICS

| Dashboard                 | Port | Subpath         | Dataset              | Repo                    |
|---------------------------|------|-----------------|----------------------|-------------------------|
| <b>strategic (PAYUNG)</b> | 8542 | /strategic/     | semua + cross-domain | strategic-dashboard     |
| ea-decision               | 8540 | /ea-decision/   | EA                   | ea-decision-dashboard   |
| dsh-analytics             | 8541 | /dsh-analytics/ | DSH                  | dsh-analytics-dashboard |
| 9router                   | 8527 | /9router-dash/  | 9router              | 9router-dashboard       |
| qdrant                    | 8529 | /qdrant-dash/   | Qdrant               | qdrant-dashboard        |
| wekan                     | 8532 | /wekan-dash/    | Wekan                | wekan-dashboard         |
| charts-admin              | 8533 | /charts-admin/  | collection engine    | collection-engine       |
| ea-dash (legacy)          | 8531 | /ea-dash/       | EA lama              | ea-dashboard            |
| dsh-dash (legacy)         | 8528 | /dsh-dash/      | DSH lama             | dsh-dashboard           |

### Strategic Dashboard — 5 Tab (payung)

- 1. **Executive Overview** — narasi sintesis: KPI inti + 3 gap auto-deteksi + sinyal + rekomendasi
- 2. **Proses Bisnis (EA)** — kematangan unit, risk map, KPI coverage
- 3. **Kebutuhan Publik (DSH)** — top query, feedback, Q&A
- 4. **Cross-Domain** — EA layanan x DSH service x query publik
- 5. **Konteks Pendukung** — biaya AI 9router (sekunder)

## L6 — KNOWLEDGE & DOKUMENTASI

| Project                              | Isi                                                              | Repo                   |
|--------------------------------------|------------------------------------------------------------------|------------------------|
| <b>ea-ugm-explorer</b>               | 6.151 diagram → markdown 28MB + SVG 78MB                         | ea-ugm-explorer        |
| <b>ugm-strategic-analysis</b>        | konsep (KONSEP-STRATEGIS.md), analisis (6 temuan, 7 rekomendasi) | ugm-strategic-analysis |
| <b>bookstack-btd</b>                 | knowledge base tim BTD                                           | bookstack-btd          |
| <b>shared-skills / hermes-skills</b> | katalog 121 skill via NAS                                        | shared-skills          |
| <b>collection-engine</b>             | charts.db + collector + admin panel                              | collection-engine      |

## L7 — INFRASTRUKTUR

| Host             | IP           | Peran                                                 |
|------------------|--------------|-------------------------------------------------------|
| NUC (def)        | 10.14.7.239  | host utama: dashboard, MCP, collector, gateway Hermes |
| PC-MONITOR (w1)  | 10.14.7.240  | worker, Nextcloud :18080, Immich                      |
| PC-AMD7900X (w2) | 10.15.17.190 | worker, 9router gateway, provider LLM                 |
| NAS Synology     | 10.15.17.194 | shared storage btdstorage                             |

- **Docker NUC:** 76+ container, bridge /24 (ipam), `sg docker -c`, port bind 0.0.0.0
- **Tailscale Funnel:** `sudo tailscale funnel --bg --set-path /<sub>/ http://localhost:<port>/<sub>/`
- **Git:** 47 repo @ dedieko-priyadi; GIT RULES (logbook commit ber-timestamp, laporan wajib URL)
- **Nextcloud:** :18080, share link baru setiap kali
- **Cron:** Hermes cron collector + hgr restart gateway (system crontab)

## SKILL TERKAIT

| Skill                       | Fungsi                                         |
|-----------------------------|------------------------------------------------|
| ugm-btd-ecosystem           | MASTER — peta 7 layer seluruh ekosistem        |
| btd-analytics-dashboard     | detail dashboard (port, pipeline, deploy pola) |
| ugm-enterprise-architecture | EA query & ekstraksi (3 jalur)                 |
| dsh-ugm-platform            | DSH platform + peta 9 project + wajah publik   |
| collection-engine           | pipeline charts.db                             |
| 9router-monitor             | monitoring 9router                             |
| server-operations           | akses 3 server + NAS                           |
| docker-container-audit      | audit container                                |
| tailscale-management        | funnel & publikasi                             |
| nextcloud-share             | file sharing                                   |
| hermes-telegram-chatbot     | pola chatbot Telegram                          |

## ALUR KERJA (saling melengkapi)

1. Data EA (eaugm\_2025) → MCP ugmcore → ea-decision-dashboard + strategic
2. Data DSH (ugm\_dsh) → MCP dsh → dsh-analytics-dashboard + strategic
3. EA × DSH → strategic Cross-Domain (gap proses vs kebutuhan publik)
4. Chatbot (ea-chatbot) → MCP ugmcore → jawab pertanyaan EA
5. Biaya AI (9router) → konteks evaluasi chatbot

- 6. Knowledge (bookstack) → dokumentasi semua project
- 7. Kanban (planka/wekan) → pelacakan project tim

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## DEPLOY POLA (dashboard baru)

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```
mkdir ~/<name>-dashboard && cd ~/<name>-dashboard
# Dockerfile: streamlit --server.port=X --server.baseUrlPath=/<subpath>
# compose: port 0.0.0.0, volume charts.db:ro, network /24
sg docker -c "docker compose up -d --build"
# user eksekusi (bubble terpisah):
sudo tailscale funnel --bg --set-path /<subpath>/ http://localhost:PORT/<subpath>/
# verify: curl funnel 200 → browser DOM → deploy-verify → commit + laporan
```

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## GIT RULES (wajib)

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1. **Deploy** → project git baru: laporan-deploy.md + README + URL lokal + funnel
2. **LOG BOOK**: tiap attempt/iterasi WAJIB commit ber-timestamp + deskripsi — git = laporan tak langsung + logbook
3. **Push** + lapor nama project + link
4. Laporan akhir WAJIB cantum URL git
5. README English, update langsung setelah fix (tanpa nunggu arahan)